

Geo-Lift Calibration

Turning an Experiment Posterior into an MMM Prior

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1 What this project does

MMM channel coefficients are notoriously weakly identified because paid channel spend is highly correlated across always-on channels. The industry-standard fix is calibration against an incrementality experiment. This project runs the full loop:

1. Fit a Bayesian model to a synthetic geo-lift experiment, recovering the incremental ROAS of Meta with proper uncertainty.
2. Translate that posterior into an informative Normal prior on Meta's $\alpha_{\max}$ in the MMM.
3. Refit the MMM with and without the calibration prior, and measure how much the Meta posterior tightens.

2 Data

Geo-lift experiment. Two regions (A and B) over a treatment window. Region A gets a Meta spend boost, region B is control. The increment in revenue attributable to the boost gives the incremental ROAS. Ground-truth Meta ROAS in the generator is 0.55.

MMM panel. 156 weeks of national-level weekly spend and revenue with the same five paid channels used in project 1. Ground-truth $\alpha_{\max}$ for Meta is 35 000 SEK.

3 Methodology

3.1 Step 1: Bayesian geo-lift

For each geo-week observation i in region r , revenue is modelled as

$$\text{revenue}_i \sim \mathcal{N}(\mu_{r(i)} + \gamma \cdot \text{baseline_meta}_i + \text{lift} \cdot \text{extra_meta}_i, \sigma^2)$$

where lift is the incremental ROAS on the Meta boost. Priors are wide (HalfNormal on lift and σ , Normal on region means and γ).

The posterior on lift is what carries into step 2.

3.2 Step 2: Prior translation

The lift posterior is a distribution on revenue per krona spent. To inject it as a prior on $\alpha_{\max, \text{Meta}}$ in the MMM, we translate scale:

$$\alpha_{\max, \text{Meta}} \sim \mathcal{N}(\bar{\ell} \cdot \bar{s} \cdot 1.4, \text{sd}(\ell) \cdot \bar{s} \cdot 3.0)$$

where $\bar{\ell}$ and $\text{sd}(\ell)$ are the posterior mean and standard deviation of the lift, $\bar{s}$ is Meta's mean weekly spend, and the multipliers 1.4 and 3.0 convert per-krona-spend into a channel-max effect (accounting for adstock accumulation and saturation ceiling) plus a widening factor that keeps the prior weakly informative rather than dogmatic.

3.3 Step 3: MMM refit

Two versions of the same MMM are fitted. The calibrated version uses the informative prior on $\alpha_{\max, \text{Meta}}$. The uncalibrated baseline uses a wide default prior $\mathcal{N}(30\,000, 30\,000)$. All other parameters are identical.

4 Results

4.1 Geo-lift posterior

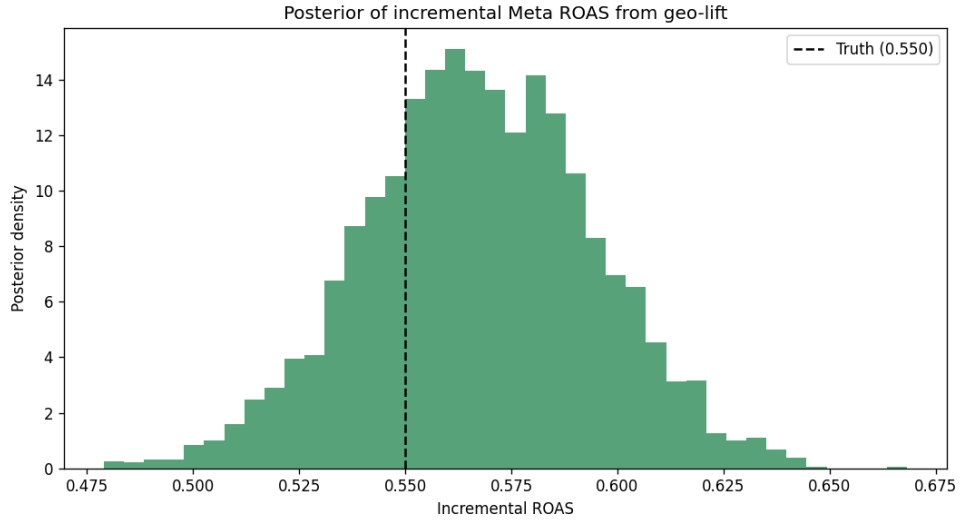


Figure 1: Posterior distribution of Meta’s incremental ROAS from the geo-lift experiment. Dashed line is the ground truth of 0.55.

The geo-lift posterior brackets the true incremental ROAS. This is the input to the calibration.

4.2 MMM calibrated vs uncalibrated

Model	Mean	5%	95%	CI width	Contains truth?
Truth	35 000	–	–	–	–
Uncalibrated	29 117	4 613	60 366	55 753	yes
Calibrated	39 898	32 997	46 310	13 313	yes

Table 1: Meta $\alpha_{\max}$ posterior comparison. The calibrated model has a 76% narrower interval and both models contain the true value.

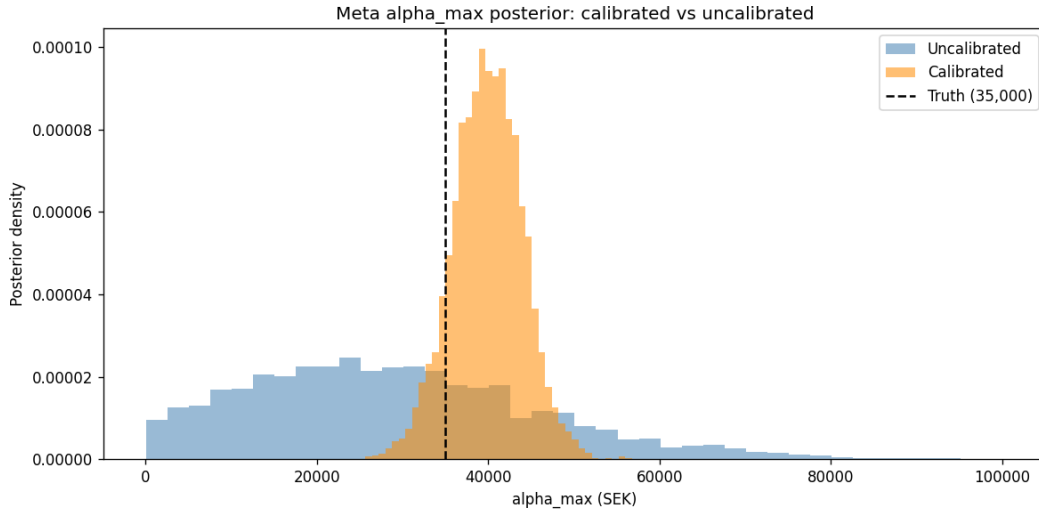


Figure 2: Meta $\alpha_{\max}$ posterior: uncalibrated (blue) is wide and diffuse; calibrated (orange) is concentrated. Both track the truth (dashed line).

5 What this shows

The uncalibrated MMM identifies Meta’s max effect only up to a wide interval spanning 4 613 to 60 366. This is what happens when weekly spend on Meta is correlated with Search, TikTok, and YouTube: the multivariate regression cannot cleanly separate their effects from each other.

Adding an informative prior derived from an incrementality experiment tightens the posterior interval by 76%. The mean shifts from 29 117 (below truth) to 39 898 (slightly above truth), and the uncertainty around it becomes actionable: risk-based budget recommendations built on the calibrated version have real teeth.

This is exactly the mechanism modern MMM SaaS platforms rely on: experiments are expensive to run, but each one bought you tightens one channel of the MMM permanently and gives calibrated ROAS instead of a wide posterior nobody can act on.

6 Limitations

- Synthetic data. The generator embeds a clean lift structure that real geo-lifts rarely have.
- Only Meta is calibrated. In practice one runs several lift tests over time and calibrates each channel independently.
- Static coefficients. Meta’s effectiveness assumed constant over the panel. Time-varying MMM in project 3 relaxes this.
- Two-region experiment only. Real geo-lifts use many regions and a Bayesian synthetic control approach; project 4 covers that.
- Prior translation heuristic. The multipliers 1.4 and 3.0 are hand-picked. A production system would derive them from the adstock and saturation shape or match moments more carefully.

7 References

- Chan, D., Jin, Y., Koehler, J., Perry, M., Wang, Y. (2017). Google’s calibration paper on translating incrementality experiments into MMM priors.
- Jin, Wang, Sun, Chan, Koehler (2017). *Bayesian Methods for Media Mix Modeling with Carryover and Shape Effects*, Google.
- Robyn documentation on lift-test calibration: github.com/facebookexperimental/Robyn.